

Experiment ‘p_4_1_a_sum_even_numbers’ Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_4_1_a_sum_even_numbers

Natural language statement: Write a method with loops that computes the sum of all even numbers between 2 and 100 (inclusive).

Method signature: p_4_1_a_sum_even_numbers() returns (sum: int)

Ensures

- sum == 2550

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method with loops that computes the sum of all even numbers between 2 and 100 (↪ inclusive).

The signature should be:

`p_4_1_a_sum_even_numbers()` returns (sum: int)

The method should respect the following contract:

ensures sum == 2550

Produce and show only the Dafny body of this method, including the curly braces that ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor ↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdxCZJR5XJlmoi5R22EGFvbYA48J

Model: gpt-5.5-2026-04-23

Created at: 1780485698

Finish reason: stop

Usage: 268 tokens in, and 638 tokens out

```
{
  sum := 0;
  var n := 1;
  while n <= 50
    invariant 1 <= n <= 51
    invariant sum == n * (n - 1)
    decreases 51 - n
  {
    sum := sum + 2 * n;
    n := n + 1;
  }
  assert n == 51;
  assert sum == 51 * 50;
  assert sum == 2550;
}
```

Final Program

```
method p_4_1_a_sum_even_numbers() returns (sum: int)
  ensures sum == 2550
{
  sum := 0;
  var n := 1;
  while n <= 50
    invariant 1 <= n <= 51
    invariant sum == n * (n - 1)
    decreases 51 - n
  {
    sum := sum + 2 * n;
    n := n + 1;
  }
  assert n == 51;
  assert sum == 51 * 50;
  assert sum == 2550;
}
```

Total Token Usage

Input tokens: 268

Output tokens: 638

Reasoning tokens: 512

Sum of 'total tokens': 906

Experiment Timings

Overall Experiment started at 1780485697460, ended at 1780485715473, lasting 18013ms (18.01 seconds)

Iteration #1 started at 1780485697465, ended at 1780485715473, lasting 18008ms (18.01 seconds)